

Model Connect App

Target for Model-based Development and Simulation for ctrlX OS
Version 4.6.1

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DC-AE/PAG-SW (MiNi)

Table of contents

1	About this documentation	4
2	Important information	4
2.1	Product information	4
3	Version 4.6.1	4
3.1	New functions	4
3.1.1	Feature 1155842: Connect to remote datalayer RT nodes	4
3.2	Resolved defects and modifications	4
3.2.1	Bug 1122422: MATLAB R2025a/b	4
3.2.2	Bug 1164008: Connected nodes could lead to automation.core shutdown	5
3.3	Notes on use and known restrictions	5
3.3.1	Bug 666836: Parameters are overwritten when opening an "External Mode" connection	5
3.3.2	Bug 954437: External mode server is disabled after it stops for the first time	5
4	Version 4.4.1	6
4.1	New functions	6
4.1.1	Feature 1058195: Support for creation of Data Layer child program nodes	6
4.2	Notes on use and known restrictions	6
4.2.1	Bug 666836: Parameters are overwritten when opening an "External Mode" connection	6
4.2.2	Bug 954437: External mode server is disabled after it stops for the first time	6
5	Service and support	7
6	Index	9

1 About this documentation

Editions of this documentation

Edition	Release date	Notes
01	2025 - 01	Model Connect App Version 4.4.1
02	2026 - 03	Model Connect App Version 4.6.1

Further information

- Refer to ➞ [Model Connect App, Application Manual](#)

2 Important information

2.1 Product information

For the latest information on our products, go to ➞ <https://www.boschrexroth.com>.

For current security advisories, please refer to ➞ <https://psirt.bosch.com/security-advisories/>

3 Version 4.6.1

3.1 New functions

3.1.1 Feature 1155842: Connect to remote datalayer RT nodes

Description

The ctrlX Data Layer has a feature called "Remote Data Layer." This allows to mirror the Data Layer of a remote ctrlX OS system in a local Data Layer.



Note: All mirrored nodes are non-realtime!

This feature makes it possible to establish a connection to the mirrored nodes, regardless of whether the original node is realtime or non-realtime.

3.2 Resolved defects and modifications

3.2.1 Bug 1122422: MATLAB R2025a/b

Description

The latest versions of MATLAB ($\geq$ R2025a) didn't support the so called "One-click external mode" solution.

Because of a change in the interface from MATLAB the deployment of a simulink model didn't run in this versions automatically.

Bug fix

The simulink models can be still generated based on the ctrlXOS_ert and ctrlXOS_grt targets. Only the deployment have to be done in this case manually, per drag and drop the model.moc file in the Model Connect frontend or over REST api.

3.2.2 Bug 1164008: Connected nodes could lead to automation.core shutdown

Description

If a connected node (which is configured as optional) isn't available at the start of a model/script and becomes available later, the classification (real-time or non realtime) and connection establishment could lead to a watchdog error.

Bug fix

The classification and connection establishment is now shifted to a non real-time thread. Therefore the execution time of the model/script isn't affected anymore.

3.3 Notes on use and known restrictions

3.3.1 Bug 666836: Parameters are overwritten when opening an "External Mode" connection

Description

When a model is transferred from Simulink to a ctrlX OS system, the model parameters are initialized with the values preset in Simulink. At this point, the default values in Simulink and the values in the model running on the ctrlX OS system in real time are identical.

If the user subsequently changes these model parameters, either directly on the Data Layer or via the "External Mode" of Simulink, these values on the ctrlX OS system no longer match the default values in Simulink. If a new "External Mode" connection is opened in case of deviating parameters, the values on the Data Layer are automatically overwritten with the configured default values in Simulink.

Workaround

After the parameters of a model have been set successfully either directly on the Data Layer or via the "Monitor & Tune" functionality, these parameters should be fed back into Simulink as new default values. After a subsequent new transfer of the model, the parameters on both sides of the "External Mode" connection match.

3.3.2 Bug 954437: External mode server is disabled after it stops for the first time

Description

When deploying Simulink models with external mode to a device running with ctrlX OS, it is only possible to connect from a Simulink client to the external mode server running on the device during the first run. After the model has stopped for the first time, the external mode server will no longer be activated again. A warning that describes this behaviour is shown on the ctrlX OS web interface, under the diagnostic (notification) bell.

Workaround

To reactivate the external mode, redeploy the Simulink model or restart the device.

4 Version 4.4.1

4.1 New functions

4.1.1 Feature 1058195: Support for creation of Data Layer child program nodes

Description

Beside the possibility of creating, reading and writing ctrlX Datalayer nodes this new functionality enables the creation of child program nodes.

This is for example necessary for commanding an axis.

4.2 Notes on use and known restrictions

4.2.1 Bug 666836: Parameters are overwritten when opening an "External Mode" connection

Description

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Workaround

To reactivate the external mode, redeploy the Simulink model or restart the device.

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Fax: **+49 9352 18 4941**

Email: [↗ service.svc@boschrexroth.de](mailto:service.svc@boschrexroth.de)

Internet: [↗ http://www.boschrexroth.com](http://www.boschrexroth.com)

Additional information on service, repair (e.g. delivery addresses) and training can be found on our internet sites.

Service worldwide

Outside Germany, please contact your local service office first. For hotline numbers, refer to the sales office addresses on the internet.

Preparing information

To be able to help you more quickly and efficiently, please have the following information ready:

- Detailed description of malfunction and circumstances
- Type plate specifications of the affected products, in particular type codes and serial numbers
- Your contact data (phone and fax number as well as your e-mail address)

6 Index

H

Helpdesk 7

Hotline 7

I

Important information 4

P

Product information 4

S

Service hotline 7

Support 7

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